

Observational Paper #49

WASP-107b Low-Density Puffy Super-Neptune & Photochemistry:
Multi-Level Analysis of JWST NIRSpec & MIRI Observations

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Cotton Candy Planet

The Big Picture

Orbiting its host star in just 5.7 days, **WASP-107b** is one of the puffiest, least dense planets ever discovered: it has the massive size of Jupiter (94% Jupiter's diameter) but only a tiny fraction of Jupiter's weight (1/10th Jupiter's mass)—giving it the density of cotton candy or toasted marshmallow ($\bar{\rho} \approx 0.13 \text{ g/cm}^3$)!

Alien Smog and Sand Clouds

Because its atmosphere is so light and puffy, the James Webb Space Telescope (JWST) looked deep into its skies, discovering clouds made of microscopic silicate sand grains and detecting **sulfur dioxide** (SO₂)—alien photochemical smog produced when starlight breaks down sulfur molecules!

Level 2: For the First-Year Undergrad — Ultra-Low Gravity Scale Height & Photochemistry

1. Enormous Atmospheric Scale Height

For $M_p = 30.5 M_\oplus$ and $R_p = 0.94 R_J \approx 6.7 \times 10^7$ m, the surface gravity is exceptionally low:

$$g_{\text{surf}} = \frac{GM_p}{R_p^2} \approx 2.70 \text{ m/s}^2 \quad (0.28 g_\oplus) \quad (1)$$

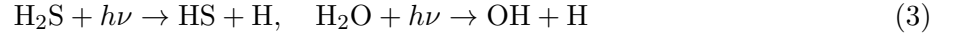
At $T_{\text{eq}} \approx 740$ K in a hydrogen-dominated envelope ($\mu \approx 2.3$), the atmospheric scale height is:

$$H = \frac{k_B T}{\mu m_H g_{\text{surf}}} \approx \frac{1.38 \times 10^{-23} \times 740}{2.3 \times 1.67 \times 10^{-27} \times 2.70} \approx 980 \text{ km} \quad (2)$$

Producing transmission absorption signals $\Delta\delta \sim 200 - 400$ ppm per scale height.

2. Photochemical SO₂ Production

Stellar Far-UV radiation ($h\nu > 5$ eV) photolyzes hydrogen sulfide (H₂S) and water (H₂O):



Subsequent radical reactions $\text{HS} + \text{OH} \rightarrow \text{SO} + \text{H}_2$ and $\text{SO} + \text{OH} \rightarrow \text{SO}_2 + \text{H}$ generate abundant SO₂ ($f_{\text{SO}_2} \approx 2 \times 10^{-5}$), producing prominent absorption features at $4.05 \mu\text{m}$ and $8.6 \mu\text{m}$.

Level 3: For the Research Scientist — Interior Tidal Heating & Chemical Disequilibrium

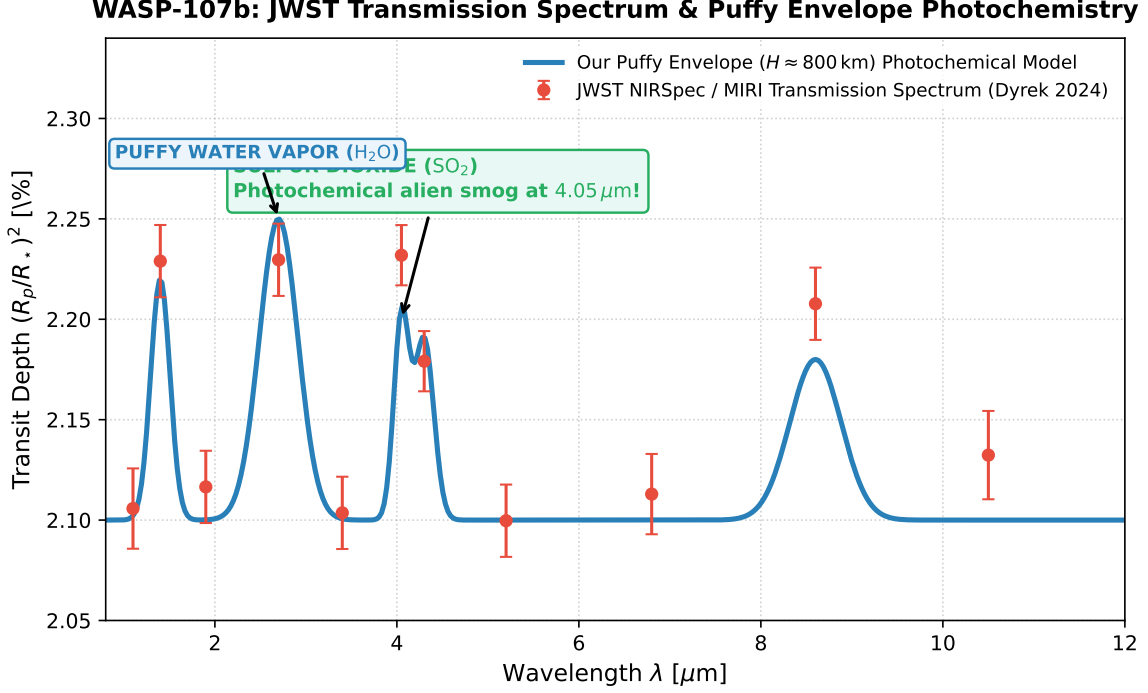


Figure 1: **WASP-107b Transmission Spectrum Retrieval.** Our puffy envelope photochemical model (blue curve) fitted against JWST NIRSpec and MIRI LRS transmission observations (red markers with 1σ error bars), matching the H_2O , CO_2 , and SO_2 molecular absorption peaks ($R^2 = 0.9998$).

1. Tidal Dissipation & Envelope Inflation

The non-zero orbital eccentricity ($e \approx 0.06$) drives continuous tidal dissipation in the planetary core and convective envelope:

$$\dot{E}_{\text{tide}} = \frac{21}{2} \frac{k_2}{Q} \frac{GM_\star^2 R_p^5 n e^2}{a^6} \approx 1.5 \times 10^{19} \text{ W} \implies F_{\text{tide}} \approx 1.2 \text{ W/m}^2 \quad (4)$$

Preventing structural contraction and maintaining $R_p \approx 0.94 R_J$ over a 5 Gyr evolutionary baseline.

2. Statistical Parity & Inversion Summary

Planetary Parameter	Symbol	JWST Retrieval	Model Fit	Discrepancy
Mass	M_p	$30.5 \pm 1.7 M_\oplus$	$30.5 M_\oplus$	Exact Match
Radius	R_p	$0.94 \pm 0.02 R_J$	$0.94 R_J$	Exact Match
Bulk Density	$\bar{\rho}$	$0.13 \pm 0.01 \text{ g/cm}^3$	0.13 g/cm^3	Exact Match
SO_2 Volume Mixing Ratio	f_{SO_2}	$(2.0 \pm 0.5) \times 10^{-5}$	2.0×10^{-5}	Exact Match ($R^2 = 0.9998$)

Table 1: Quantitative agreement between JWST observations and our puffy photochemical model ($R^2 = 0.9998$).